## Experiment -7

## Python Program To Implement Breadth-First Search

**Code:**

```
from collections import deque

def bfs(graph, start):
    visited = set()
    queue = deque([start])

    while queue:
        node = queue.popleft()

        if node not in visited:
            print(node, end=" ")
            visited.add(node)

            for neighbor in graph[node]:
                if neighbor not in visited:
                    queue.append(neighbor)

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
```

}

print("BFS Traversal:")

bfs(graph, 'A')

```python
from collections import deque

def bfs(graph, start):
    visited = set()
    queue = deque([start])

    while queue:
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        if node not in visited:
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}

print("BFS Traversal:")
bfs(graph, 'A')
```

```
BFS Traversal:
A B C D E F

...Program finished with exit code 0
Press ENTER to exit console.
```